Adaptive Spectral–Spatial Multiscale Contextual Feature Extraction for Hyperspectral Image Classification

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***Abstract*— In this article, we propose an end-to-end adaptive spectral–spatial multiscale network to extract multiscale contextual information for hyperspectral image (HSI) classification, which contains spectral feature extraction (FE) and spatial FE subnetworks. In spectral FE aspect, different from previous methods where features are obtained in a single scale, which limits the accuracy improvement, we propose two schemes based on band grouping strategy, and the long short-time memory (LSTM) model is used for perceiving spectral multiscale information. In spatial subnetwork, on the foundation of existing multiscale architecture, the spatial contextual features which are usually ignored by previous literature are successfully obtained under the aid of convolutional LSTM (ConvLSTM) model. Besides, a new spatial grouping strategy is proposed for convenience of ConvLSTM to extract the more discriminative features. Then, a novel adaptive feature combining way is proposed considering the different importance of spectral and spatial parts. Experiments on three public data sets in HSI community demonstrate that our methods achieve competitive results compared with other state-of-the-art methods.**

***Index Terms*— Adaptive, convolutional long short-time memory (ConvLSTM), end-to-end, hyperspectral image (HSI) classification, multiscale contextual information.**

I. Introduction

As a result of the rapid development of satellite sensor technology, hundreds of continuous and narrow spectral bands are now available for Earth observation. Abundant spectral information contained in a hyperspectral image (HSI) can precisely characterize the physical properties and bring great convenience for land objects recognition. There are many important applications including precision agriculture [1], object detection [2], and environmental monitoring [3]. Among these fields, the land cover classification, which aims to assign a certain semantic label for pixels on the whole image with the help of the unique spectral and spatial characteristics of HSI, has always been a hot and important topic over the past years.

The traditional classification methods such as k-nearest neighbors [4], logistic regression (LR) [5] tend to regard the original pixel vector as the input feature. However, since there are a large amount of channels in HSI, the classifier needs to process these features in high-dimensional space. The unbalance between the limited samples and high-dimensional features challenges the classifier and makes it difficult to distinguish the sample points effectively. This problem is called the curse of dimensionality or Hughes phenomenon, which usually causes overfitting. To address this problem, dimensionality reduction (DR) is often treated as a crucial step in the preprocessing procedure. Typical DR operation includes feature selection (FS) and feature extraction (FE). FS aims to preserve the most valuable and discriminative channels of the original band set while discarding the bands that have no representation ability with some specially designed criteria [6]. Different from FS, FE can transform the vectors in HSI from the original high-dimensional space to low-dimensional one with a suitable mapping, and the projected vectors often have a higher separability. Some commonly used FE methods include principal component analysis (PCA) [7], independent component analysis [8], and linear discriminant analysis [9].

Besides the spectral features, a lot of recent works have considered the spatial information. Among these methods, support vector machine (SVM) is conducted as the most commonly used classifier owing to its stable performance when dealing with high-dimensional data. For example, several kinds of morphological profiles are used to fuse the spectral and spatial features through opening and closing operations, the obtained features are then fed into a SVM classifier [10], [11]. Gu *et al.* [12] and Fang *et al.* [13] improved the SVM classifier performance by adopting the multiple kernel learning strategy. In addition, superpixel is developed to simplify the classification complexity [14]–[16]. For instance, the original HSI is integrated with multiscale superpixel

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segmentation maps for subspace-based SVM [14]. In [15], the spectral and spatial information based on superpixel segmentation are both fed into the extended random walker framework, although the SVM is used to yield the prior probabilities of the superpixels. These aforementioned methods tend to conduct feature engineering with the help of spectral–spatial information on the HSI and then generate the classification map. There are also some methods that give other solutions to this issue. They refine the previous classification results with spatial contextual information by means of a Markov random field regularization [17]. Additionally, some other approaches such as loopy belief propagation [18], sparse representation [19], [20], and Gabor filter [21] have also been employed.

However, traditional feature engineering, as the methods mentioned above, usually produces the handcrafted features, which should be considered as the shallow feature because these aforementioned methods belong to shallow-layer models. For example, linear SVM and LR can be viewed as a single-layer classifier, whereas kernel SVM or decision tree are thought to have two layers [22]. These shallow-layer features often cannot reflect the essential characteristics of the observed objects, and they are not robust enough in complex circumstances [23]. In recent years, with the hierarchical and representation learning characteristics, deep learning technologies have drawn the increasing attention of public. The literal meaning of deep learning is the stacking of neurons, and then constructing a neural network which has so many layers that can be called "deep." This design lets current extracted features strongly depend on previous layers. As the layers of network deepen, the constant linear or nonlinear transformations make the features more semantic, abstract, and stable, which are still robust even if the environment changes dramatically. These features are closer to the intrinsic properties of the identified object compared to shallow features. Unlike traditional methods, such as SVM which receives the designed handcrafted features, the deep neural networks (DNNs) can learn representative features without expert knowledge.

Deep learning methods have shown incredible breakthrough in many computer vision tasks, such as classification [24]–[26], detection [27], [28], and segmentation [29], and have been applied to HSI classification field widely [30]. For example, a typical DNN, stacked autoencoder is used to separately extract spectral and spatial features with layerwise stacking pretraining strategy [31], and this strategy ensures the network deep enough so as to obtain the deep features. In addition, another kind of DNN called deep belief network, has also been employed [32]. However, since a large amount of parameters needs to be optimized, the training of DNNs is time-consuming and requires sufficient computing resources. Recently, with the local perception and parameter sharing characteristics, convolutional neural networks (CNNs) have demonstrated their effectiveness in HSI classification [33]–[40]. For example, in [33], the spectral vectors are fed into an architecture with five layers, including convolutional layer (Conv), max pooling layer, and fully connected layer (FC). Li *et al.* [34] used CNN as a feature extractor to obtain discriminative representations from designed pixel pairs and then determined the final label with the voting strategy. In [35] balanced local discriminant embedding algorithm and CNN are employed, respectively, to extract spectral and spatial features, which are then fused together for HSI classification. In addition, 3-D convolution is used in [38] and [39] to extract spectral–spatial joint features. However, classification accuracy is still limited since the extracted features of the aforementioned methods are all in a single scale. Therefore, [36] and [37] obtain multiscale features by combining both shallow and deep convolutional features in order to improve the classification quality.

All the aforementioned methods regard pixels in HSI as vectors, meaning each pixel represents a data point in d-dimensional space where d is the number of bands. However, the hundreds of bands in HSI are continuous and narrow, which have strong internal correlation. Thus, these bands should be considered as sequences instead of vectors. In natural language processing field, the recurrent neural networks (RNNs) and their extensions have been used widely because of their excellent sequence modeling ability [41]–[43]. RNN and its variants including long short-term memory (LSTM) and gated recurrent units (GRUs) can capture the temporal variability of long sentences in natural language. It is natural to apply these models on HSI, because pixels can also be regarded as the sequential data and reflect the spectral variability that the reflectance changes with wavelength. These models have been involved in a large amount of works. For example, LSTM and RNN with T_b time steps are used in [44] and [45] to classify the spectral vector directly, where T_b is the number of bands in HSI. Xu *et al.* [37] split the spectral vector into several parts in the direction of channels, and fed them into LSTM in order, reducing the difficulty of extracting information for LSTM. Cascaded-RNN is employed in [46] to further extract information among RNN layers. In the aspect of spatial information extraction, it is unsuitable to feed the multidimensional spatial patches into these sequential models directly, since these models need 1-D input. To tackle this problem, Feng *et al.* [44] split the patches of the first principal component in row direction, and fed these rows into LSTM, while RNN and GRU are separately used in [46] and [47] to extract spatial information from features that are obtained with CNN and FC layers.

However, the original spatial structure in HSI is inevitably lost when transforming the multidimensional data to 1-D. In fact, the size is limited because of the FC layers in these sequential models. In order to better extract the spatial sequential information while preserving the shape of input data, Shi *et al.* [48] developed the convolutional LSTM (ConvLSTM) on the foundation of LSTM by replacing the FC layer with convolutional layer in the gates. There are also some works that apply ConvLSTM in HSI classification [49]–[51]. For example, ConvLSTM combined with 2-D or 3-D convolution is proposed in [49] in order to capture the relation of different channels in the feature maps at CNN. The bidirectional ConvLSTM is also used in [50] to extract spectral–spatial features for HSI classification.

It can be observed that almost all the methods mentioned above except [37] ignore the issue of scale because the last

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feature extracted for classification is always in a single scale, meaning the pixels of these features all have the same receptive field, which restricts the classification accuracy. However, the spectral part in [37] is still a single-scale model. Inspired by the MSCNN in [37], it is natural to conduct multiscale sequential learning, so as to perceive and generate more abundant information, which may be favorable for classification. In addition, the aforementioned methods usually cannot conduct multiscale FE and spatial relation information perception simultaneously. For example, 3-D CNN [38], [39] and MSCNN operate the feature maps by sliding the convolutional kernels, which are unable to obtain the spatial contextual information of current scale.

To tackle these aforementioned problems, in this article, an adaptive spectral–spatial multiscale network (ASSMN) is proposed for HSI classification. It contains two subnetworks including spectral multiscale network (SeMN) and spatial multiscale network (SaMN) for spectral and spatial information perception, respectively. "Adaptive" means the extracted features of these two parts are combined reasonably and parameters are automatically determined via an end-to-end learning strategy. By sharing a uniform loss function, these two parts are incorporated into a unified framework and trained simultaneously through backpropagation. In the perspective of spectral information extraction, multiscale spectral variability is perceived with the flexible using of LSTM. Inspired by Hu *et al.* [49] and Liu *et al.* [50], ConvLSTM is implemented in spatial information perception network in order to achieve multiscale FE and contextual information capturing. The major contributions of this article are summarized as follows.

1) An end-to-end adaptive classification network, namely ASSMN, is proposed. Compared with existing spectral–spatial joint networks, our model can not only fully take advantage of multiscale features but also have the capability of adaptive learning so as to combine spectral and spatial features more effectively.
2) In order to learn more discriminative features, two multiscale schemes are designed for spectral information extraction. Scheme1 is operated on original spectral vectors, while another one is based on convolutional features. Two corresponding networks are further developed using these two schemes.
3) A novel spatial information extraction network based on ConvLSTM is proposed, which achieves multiscale feature perception and contextual information capturing simultaneously.
4) The proposed networks yield satisfactory performance in three benchmark data sets that are commonly used in HSI classification community. The quantitative results and classification maps show the competitiveness of the proposed framework compared with other state-of-the-art methods.

The remainder of this article is organized as follows. Section II gives a brief introduction of LSTM and ConvLSTM. Section III describes the proposed spectral FE networks, along with two multiscale schemes in detail. Sections IV and V introduce the structure of the proposed spatial subnetwork and ASSMN, respectively. Experiments and related comprehensive analysis are presented in Section VI. Finally, Section VII summarizes this article.

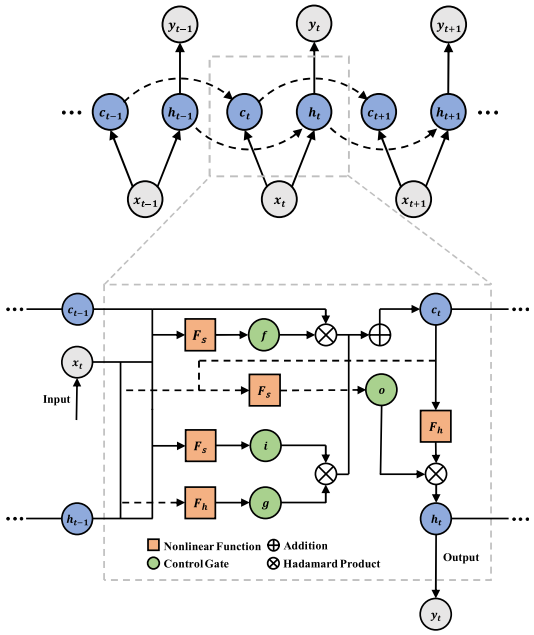

Fig. 1. LSTM structure.

II. Related Work

In this section, we will provide a brief introduction of LSTM and ConvLSTM, which are also the crucial technologies of the proposed methods.

A. LSTM

Although RNN has been applied into many sequential modeling applications, it still has potential risks of gradient vanishing or exploding in backpropagation procedure when dealing with long-term data [52]. Thus, LSTM is proposed in [53]. With the gating mechanism and memory cell, LSTM can effectively capture the contextual information of adjacent data in sequences. The structure of LSTM is shown in Fig. 1. It can be observed that the LSTM cell constantly updates the hidden value h and cell value c with the help of three control gates, including input gate i, forget gate f, and output gate o, which are used to discard, retain, or amplify signal so as to achieve information control and transformation. The calculation process in one LSTM cell at time t is

$$\begin{aligned} i &= F_s(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci} \otimes c_{t-1} + b_i) \\ f &= F_s(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf} \otimes c_{t-1} + b_f) \\ g &= F_h(W_{xg}x_t + W_{hg}h_{t-1} + b_g) \\ c_t &= i \otimes g + f \otimes c_{t-1} \\ o &= F_s(W_{xo}x_t + W_{ho}h_{t-1} + W_{co} \otimes c_t + b_o) \\ h_t &= o \otimes F_h(c_t) \end{aligned} \tag{1}$$

where x_t, h_t, and c_t represent the input, hidden state, and cell state at the tth time step, respectively. $\{(W_{\odot i}, b_i), (W_{\odot f}, b_f), (W_{\odot o}, b_o), \odot = x, h, c\}$ denotes the

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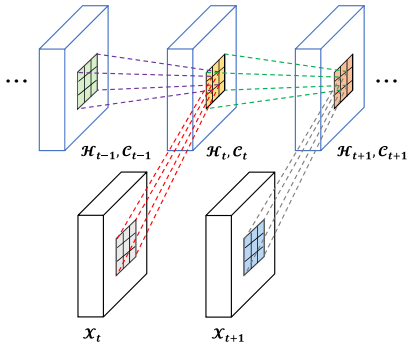

Fig. 2. Illustration of ConvLSTM.

weight matrices and bias vectors of input gate i, forget gate f, and output gate o, whereas $\{(W_{\odot g}, b_g), \odot = x, h\}$ is the weight and bias of transition value g. $F(\cdot)$ is used for nonlinear mapping. Usually sigmoid function $s(x) = 1/(1 + \exp(-x))$ and hyperbolic tangent function $\tanh(x) = (\exp(x) - \exp(-x))/(\exp(x) + \exp(-x))$ are employed and we denote them with F_s and F_h, respectively. "⊗" is the hadamard product, meaning pixelwise multiplication.

LSTM can well-capture contextual information existing in the adjacent bands in spectral domain. But it is still limited for spatial contextual information extraction. Although Feng *et al.* [44] had attempted to obtain spatial information with LSTM, the original image has to be flattened into an 1-D vector, and the intrinsic spatial structure are inevitably lost, due to the restriction of input shape.

B. Convolutional LSTM

The aforementioned weakness in LSTM is mainly originated from the FC layers, which restrict the input data shape. Recently, convolution operation has been widely applied for spatial encoding in HSI classification community owing to the local perception ability, which also preserves the spatial relationship among adjacent pixels. Thus, it is natural to combine the convolution filter and LSTM to extract contextual information of multidimensional data. This idea has been implemented by Shi *et al.* [48] by constructing ConvLSTM for precipitation prediction.

Fig. 2 shows the architecture of ConvLSTM. Compared with LSTM, the input, hidden, and memory state in ConvLSTM are all multidimensional data, and this procedure is supported by convolution kernel. Therefore, ConvLSTM can model variability in spatial sequence data. The calculation of ConvLSTM in tth timestep can be formulated as

$$\begin{aligned} i &= F_s(W_{xi} * \mathcal{X}_t + W_{hi} * \mathcal{H}_{t-1} + W_{ci} \otimes \mathcal{C}_{t-1} + b_i) \\ f &= F_s(W_{xf} * \mathcal{X}_t + W_{hf} * \mathcal{H}_{t-1} + W_{cf} \otimes \mathcal{C}_{t-1} + b_f) \\ g &= F_h(W_{xg} * \mathcal{X}_t + W_{hg} * \mathcal{H}_{t-1} + b_g) \\ \mathcal{C}_t &= i \otimes g + f \otimes \mathcal{C}_{t-1} \\ o &= F_s(W_{xo} * \mathcal{X}_t + W_{ho} * \mathcal{H}_{t-1} + W_{co} \otimes \mathcal{C}_t + b_o) \\ \mathcal{H}_t &= o \otimes F_h(\mathcal{C}_t) \end{aligned} \tag{2}$$

where $\mathcal{X}_t, \mathcal{H}_t$, and $\mathcal{C}_t$, respectively, denote the multidimensional input, hidden state, and cell state at time t. "$*$" is the convolution operation, while other variables and symbols have the similar meaning with the standard LSTM. It is obvious that ConvLSTM substitutes the original matrix multiplication with convolution and maintains the gate control mechanism, which provides ability of data communication and transformation in inter and intralayers. Thus, ConvLSTM can be used for modeling variability of multidimensional data, especially for spatial information.

In this article, the LSTM is adopted in our spectral subnetwork to extract sequential information among neighbor bands, whereas ConvLSTM is flexibly employed in the spatial part for contextual information extraction.

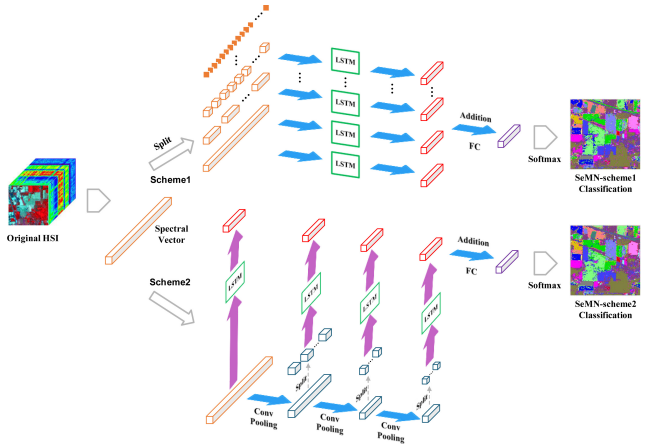

Fig. 3. Proposed SeMN, which contains two paths representing two schemes, respectively. Split means dividing the vector to several parts as inputs at different time steps for LSTM. Conv, pooling, and FC represent convolutional layer, max pooling layer, and fully connected layer.

III. Spectral Multiscale Subnetwork

Although previous jobs such as [37] and [44]–[46] have applied sequential model to acquire spectral information, the issue of scale is ignored by them and the extracted feature for classification is only in a single field of vision. However, the contextual information at different scales is also considered having a contribution for improving classification quality. To this end, the relation of different reflectances in a wide or narrow range of wavelength is also exploited for land cover recognition in the proposed method.

After the processing of normalization, the original spectral vector can be fed into the SeMN. The simplest way to explore the effect of multiscale is diversifying the scale of input data for LSTM. We propose two schemes to achieve multiscale effect from perspective of wavelength and receptive field. The proposed SeMN and these schemes are shown in Fig. 3.

A. Scheme1: Multiscale of Wavelength

In early LSTM-based spectral classification methods, the reflectance value in each band of current pixel is fed into LSTM in turn at every time step [44]. However, this manner may lead to the phenomenon of long-term dependence [53], bringing about the problems of gradient vanishing or exploding due to the hundreds of bands in HSI. One possible solution

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is to use more bands at a time by grouping spectral vectors. Xu *et al.* [37] presented two novel grouping strategies. However, these strategies are operated at a single scale, which limits the performance of LSTM.

A natural idea to obtain multiple spectral segments which locate at different wavelength range is to repeatedly use the grouping strategy several times, that is what the scheme1 in Fig. 3 means. The simplest way is to constantly halve the spectral vector, and feed the results of into different LSTMs. Then the final spectral feature is obtained by merging the outputs of all LSTMs. For example, an LSTM with two time steps is used for the first bisection spectral vector. After halving these two parts again, a spectrum with four segments will be fed into another LSTM and features at current scale can be obtained. The whole spectral vector is also directly utilized in another LSTM for global information acquisition. Finally, features from all LSTMs are further merged by elementwise addition.

B. Scheme2: Multiscale of Vision Field

With the consideration of alternate grouping, Xu *et al.* [37] further improved the discrimination of extracted features. The essence is to reduce the distance between the inputs at adjacent time step in feature space, making the transitions of these vectors more natural. Benefitting from this thinking, scheme2 aims to generate sequential data with a closer distance between adjacent time steps. This scheme can achieve multiscale FE while ensuring the coherence of the input sequence. As shown in the lower path of Fig. 3, by means of multiple convolutional layers, several feature maps are obtained. In fact, these feature maps in different layers possess multiscale information because deep networks have the characteristic of hierarchical learning, and the receptive field in deeper layers is larger than the one in shallow layers. There must exist intersection in the vision fields of adjacent channels within each group if the stride in convolution is smaller than kernel size. After obtaining these feature maps, similar to the split operation in scheme1, these feature maps are divided into several parts and fed into different LSTMs separately, while each feature map is split only one time. The hyperparameters in scheme2 that need to be adjusted are the time step numbers of different LSTMs on each feature map. In other words, different feature maps may be divided into different number of groups. These parameters are discussed in Sections VI-C1 and VI-C2. Moreover, the global information and addition operation are also employed in this scheme.

Finally, a FC layer and softmax function are used in SeMN for transforming the fused features of these two schemes to classification map.

IV. Spatial Multiscale Subnetwork

Fig. 4 demonstrates the work flow of the proposed SaMN. First, PCA decomposition is used on original HSI for dimension reduction. After the processing of normalization, spatial patches with a size of $H \times W \times C$ around the target center pixel are obtained and can be fed into the network, where H, W, and C are the height, weight, and the channel number

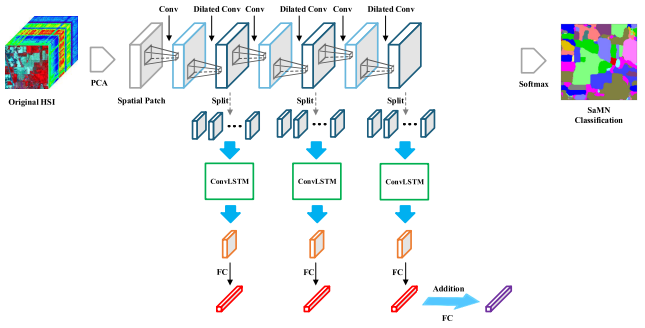

Fig. 4. Proposed spatial classification subnetwork SaMN. After PCA decomposition, the spatial patches are fed into the network. Conv, Dilated Conv, and FC mean convolutional layer, dilated convolutional layer, and fully connected layer, respectively. Split represents dividing the feature map to several parts as inputs at different time steps for ConvLSTM.

after dimension reduction of these patches. In this article, H is set equal to W for convenience.

MSCNN proposed in [37] provides an example for multiscale spatial information extraction. However, this structure still needs to be improved. Specifically, the pooling layer in MSCNN is mainly used for increasing the receptive field so as to obtain the multiscale features. It is unavoidable to reduce the size of feature map and lose spatial information. To this end, we substitute all pooling layers with dilated convolutional layers [54]–[57] and followed by rectified linear unit (ReLU) activation. The dilated convolution solves this problem by embedding holes in convolutional kernels, increasing the receptive field while preserving the resolution of feature map in the meantime. The ReLU activation is used to achieve the nonlinear mapping in original max pooling layer. In addition, the spatial contextual information is also considered in SaMN, making up the weakness of MSCNN that the relational information cannot be perceived.

As shown in Fig. 4, additional spatial contextual information extraction branches are appended to the dilated convolutional layers, which are similar to the multiscale information collection structure in MSCNN. The purpose of our design is to obtain spatial sequential information while guaranteeing hierarchical learning in backbone branch. Thus, the extracted features also have multiscale characteristics, which need to be split and divided into several parts in spatial dimensions. As for the $H \times W \times C$ patches shown in Fig. 5, suppose the data fed into the ConvLSTM at each time step is with a size of $s \times s \times C$, meaning there are H/s and W/s time steps in row and column directions, respectively. In this way, we can obtain H/s and W/s groups in these two directions, resulting in $H \cdot W/s^2$ groups from each patch. To achieve the goal, a natural idea is to first group pixels in order at one direction and then repeat the process in another direction. The corresponding result is shown in Fig. 5(a). However, unlike spectra obtained in a narrow wavelength range that have a strong correlation between each other, spatial regions may present weak relationship and fail to possess significant contextual information if they are not adjoined especially in a large scale area due to the limitation of spatial resolution in HSI. For example, it is difficult for a computer to identify

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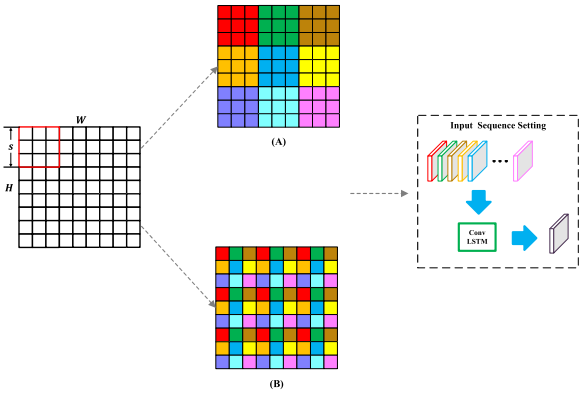

Fig. 5. Two typical spatial grouping strategies. (a) Traditional grouping strategy and (b) proposed strategy. Pixels provided for ConvLSTM in the same time step are marked with the same color.

the relationship between red part and brown part in Fig. 5(a) because they are far away from each other.

Inspired by 1-D alternate grouping strategy proposed in [37] for processing spectral vectors, we extend this idea to multidimensional level and aim to group the spatial patches for ConvLSTM by successively execute this strategy in row and column directions, the related diagram is depicted as Fig. 5(b). It can be seen that the proposed strategy enables the inputs of different time steps to possess better communication because almost all pixels except borders are adjoined with pixels from other time steps, indicating a closer distribution in feature space of different time step data and making it more convenient for ConvLSTM to extract adjacent information. In addition, compared with strategy A, the proposed strategy B ensures that data in all time steps have a larger coverage, and more conductive to perceive global information for ConvLSTM.

After grouping process, the obtained spatial sequential data can be fed into ConvLSTM. The general approach is providing these grouped data into a ConvLSTM in turn, as the plain style in Fig. 6 shows, indicating that the time steps are equal to the number of groups. However, this way ignores the intrinsic spatial structure and only one direction spatial contextual information can be detected by ConvLSTM. In other words, sequential information in another direction is lost. For example, ConvLSTM cannot effectively extract the relational feature between red part and orange part of Fig. 5(a) and (b) at row direction if the sequences are input into ConvLSTM in plain style shown in Fig. 6. In order to obtain contextual information in these two directions, the plain style solution can be directly and separately applied in these two directions, but it will inevitably increase computing burden because a large amount of data are repeatedly used. Furthermore, the long-term data may cause gradient vanishing or exploding, and have a risk of overfitting.

In order to solve this problem, we propose a solution called cascade style which balances the computational efficiency and relational information perception in two directions, as shown in Fig. 6. This solution first captures sequential information using H/s ConvLSTMs which have W/s time steps and H/s

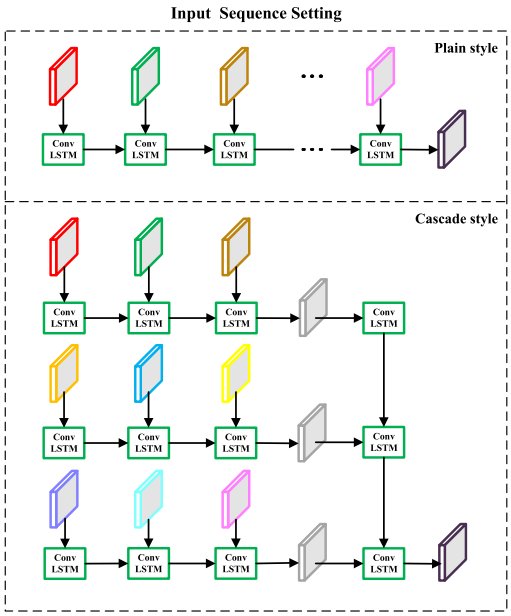

Fig. 6. Two input sequence settings: plain style is the traditional way that directly feeds these split patches into ConvLSTM in turn, and the lower part is the proposed cascade style which extract features by a hierarchical structure.

contextual features are obtained if there are H/s groups in row direction. The extracted features represent the global contextual information in these H/s sequences in column direction. Then the relevance of these H/s contextual features in column direction are further extracted by an another ConvLSTM with H/s time steps in row direction. Through this cascade operation, whole contextual information in the original input spatial patch is gradually gained. Its essence is in obtaining larger range features based on previous smaller parts using a hierarchical architecture.

The subsequent steps are the same as MSCNN. The extracted spatial contextual features in all branches will be flattened and fed into the corresponding FC layers. Results in each branch are combined with an addition function. The final classification map is obtained after an another FC layer and a softmax function.

However, there is a large amount of linear or nonlinear calculation in ConvLSTM, causing the complex data transformation which has a risk of overfitting. Therefore, it is necessary to carry out regularization constraints. Batch normalization (BN) [58] and dropout [59] are widely used regularizers and they have been applied in the proposed SaMN. The dropout is used on the flattened features for accelerating calculation and training randomization. While BN layer is embedded behind the convolutional layer in ConvLSTM of all branches instead of backbone network, since the computational processes mainly exist in ConvLSTM. Additionally, all hyperbolic tangent functions F_h in ConvLSTM are also replaced by ReLU so as to provide a simpler mapping and avoid overfitting.

V. ASSMN

After obtaining the multiscale spectral and spatial contextual features which are separately extracted through aforemen-

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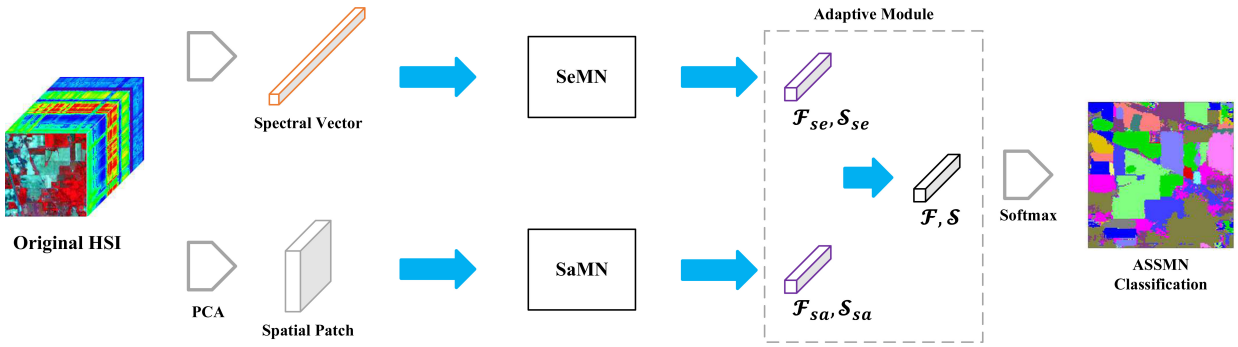

Fig. 7. Proposed framework ASSMN, including spectral and spatial subnetworks SeMN, SaMN. The extracted features are integrated with an adaptive module to generate the final classification map.

tioned subnetworks SeMN and SaMN, an adaptive spectral–spatial joint classification network ASSMN is constructed on the foundation of these features. The key issue here is how to effectively combine these features. "Adaptive" means this combining procedure is operated automatically. Fig. 7 displays the proposed ASSMN and the obtained features in fact are transition values after addition and FC layers in SeMN and SaMN. Moreover, these features have been marked in purple in Figs. 3 and 4. In these features, $\mathcal{F}_{\text{se}}$ and $\mathcal{F}_{\text{sa}}$ are the extracted features from SeMN and SaMN, while $\mathcal{S}_{\text{se}}$ and $\mathcal{S}_{\text{sa}}$ are the corresponding score vectors whose length is equal to the number of categories (before softmax function). The value of each neuron reflects the importance of the corresponding classes. Usually, the joint score vector $\mathcal{S}$ is obtained in the following ways.

- *Feature Concatenation:* This way is used in [37], the joint feature $\mathcal{F}$ is acquired by directly concatenating $\mathcal{F}_{se}$ and $\mathcal{F}_{sa}$ in channel dimension, and $\mathcal{S}$ is obtained through an FC layer

$$\begin{aligned} \mathcal{F} &= \text{Concat}(\mathcal{F}_{se}, \mathcal{F}_{sa}) \\ \mathcal{S} &= W\mathcal{F} + b \end{aligned} \tag{3}$$

  where W, b are the weight matrix and bias of the FC layer (same below).
- *Feature Weighted Addition:* This method is proposed in [60], the joint feature $\mathcal{F}$ is acquired by learning a weight parameter m, and the score vector $\mathcal{S}$ is obtained through the same way with feature concatenation

$$\begin{aligned} \mathcal{F} &= F_s(m) \cdot \mathcal{F}_{se} + (1 - F_s(m)) \cdot \mathcal{F}_{sa} \\ \mathcal{S} &= W\mathcal{F} + b \end{aligned} \tag{4}$$

  where the sigmoid function F_s is used to convert the weight parameter m to the range of (0,1). In fact, feature addition can be seen a special case of this way where weights on both sides are equivalent.

However, it seems that the feature weighted addition is unable to perform better than feature concatenation. This conclusion can be simply proved as follows: the feature concatenation procedure can be expanded as:

$$\begin{aligned} \mathcal{S} &= W\mathcal{F} + b \\ &= W_1\mathcal{F}_{se} + W_2\mathcal{F}_{sa} + b_1 + b_2 \\ &= A_1 + b_1 + A_2 + b_2 \end{aligned} \tag{5}$$

where $W = \text{Concat}(W_1, W_2), b = b_1 + b_2$, W_1, W_2 are two different matrices which can be learned automatically, while feature weighted addition is

$$\begin{aligned} \mathcal{S} &= W\mathcal{F} + b \\ &= F_s(m) \cdot W\mathcal{F}_{se} + (1 - F_s(m)) \cdot W\mathcal{F}_{sa} + b. \end{aligned} \tag{6}$$

Formula (5) and (6) show that feature weighted addition is very similar to feature concatenation, or maybe even worse than the latter way, because two different features $\mathcal{F}_{\text{se}}$ and $\mathcal{F}_{\text{sa}}$ are transformed through the same matrix W instead of two corresponding learned matrices W_1 and W_2, meaning the obtained fusion feature $\mathcal{F}$ may have a worse representing ability compared with the situation of feature concatenation.

Considering the aforementioned issue and carrying on the thinking of different importance in spectral and spatial aspects, we propose to use "score weighted" instead of "feature weighted." It can be simply understood that the joint score vector $\mathcal{S}$ is obtained by weighting $\mathcal{S}_{\text{se}}$ and $\mathcal{S}_{\text{sa}}$, and this procedure can be represented as

$$\begin{aligned} \mathcal{S} &= F_s(m)\mathcal{S}_{se} + (1 - F_s(m))\mathcal{S}_{sa} \\ &= F_s(m) \cdot W_{se}\mathcal{F}_{se} + (1 - F_s(m)) \cdot W_{sa}\mathcal{F}_{sa} \\ &\quad + F_s(m) \cdot b_{se} + (1 - F_s(m)) \cdot b_{sa} \\ &= \lambda_1(A_1 + b_1) + \lambda_2(A_2 + b_2) \end{aligned} \tag{7}$$

where $\lambda_1 + \lambda_2 = 1$, $W_{\text{se}}, W_{\text{sa}}, b_{\text{se}}, b_{\text{sa}}$ are different transformation matrices or bias vectors of $\mathcal{F}_{\text{se}}, \mathcal{F}_{\text{sa}}$. A_1 and A_2 are the matrix products of generalized spectral or spatial transformation matrices [W_1, W_2 in formula (5) or $W_{\text{se}}, W_{\text{sa}}$ in formula (7)] and $\mathcal{F}_{se}, \mathcal{F}_{sa}$. Compared to formulas (5) and (6), it can be seen that the proposed method not only preserves the two different matrices but also emphasizes the different importance of these two parts, and all parameters can be learned in network training so as to ascertain the proportions of these two parts adaptively. It can be regarded as an advanced version of feature concatenation, thus empowering model a stronger data representation ability than feature weighted addition.

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Finally, a multitask cross entropy loss is utilized to optimize the proposed ASSMN, which can be defined as

$$L = L_{\mathcal{S}} + L_{\mathcal{S}_{se}} + L_{\mathcal{S}_{sa}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{M} y_{ic}\left[\log\left(z_{ic}^{\mathcal{S}}\right)+\log\left(z_{ic}^{\mathcal{S}_{se}}\right)+\log\left(z_{ic}^{\mathcal{S}_{sa}}\right)\right]. \quad (8)$$

Here, N is the number of training samples, M is the class number, and y_{ic} is the indicator variable

$$y_{ic} = \begin{cases} 1, & c = y_i \\ 0, & c \neq y_i \end{cases} \quad (9)$$

where y_i is the truth label. $L_\bullet$ and $z_{ic}^{\bullet}$ separately represent loss and probability vector ($\bullet = \mathcal{S}, \mathcal{S}_{\mathrm{se}}, \mathcal{S}_{\mathrm{sa}}$), which can be computed through softmax function. For example, $z_{\mathrm{ic}}^{S} = (\exp(\mathcal{S}_{\mathrm{ic}}))/(\sum_{c=1}^{M}\exp(\mathcal{S}_{\mathrm{ic}}))$.

VI. Experimental Results

In this section, a series of qualitative and quantitative comprehensive analyses of the proposed method are conducted on three public HSI classification data sets, including Indian Pines, Pavia University, and Kennedy Space Center (KSC) data sets.

A. Data Sets Description

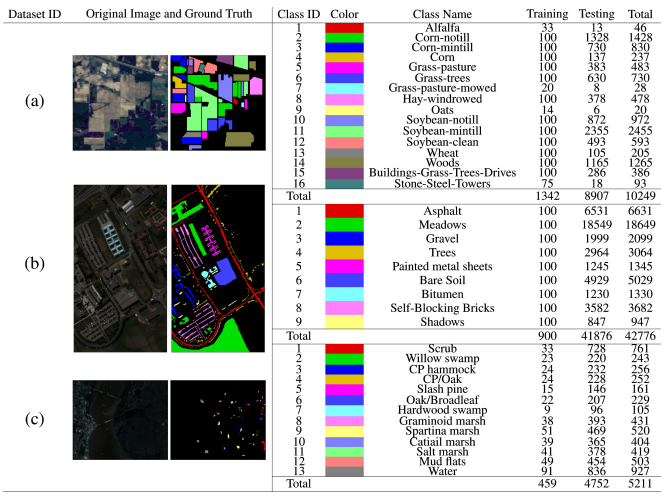
1) Indian Pines: This data set was gathered at Northwestern Indiana by Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1992, consisting 200 bands with size of 145 × 145 pixels that in 20-m spatial resolution after water absorption bands were removed and in the wavelength range of 0.4–2.5 μm. 16 vegetation classes are involved in this scene, related information, such as original image visualization, ground truth, configuration of training and testing samples, has been presented in Fig. 8(a).

2) Pavia University: This data set was obtained over Pavia University at Northern Italy by Reflective Optics System Imaging Spectrometer (ROSIS) in 2001, consisting 103 bands with size of 610 × 340 pixels that in 1.3-m spatial resolution and in the wavelength range of 0.43–0.86 μm. Nine categories are included in this data, which is shown in Fig. 8(b).

3) KSC Data Set: This scene was collected at KSC, Florida by AVIRIS in 1996. After removing water absorption and low SNR bands, 176 bands in range of 0.4–2.5 μm are remained. 512 × 614 pixels are included in this data set and in spatial resolution of 18 m. 13 classes representing the various land cover types that occur in this environment were defined for classification. Details of this data set are depicted as Fig. 8(c).

B. Model and Experimental Setting

Before the discussion of quantitative analysis, it is necessary to introduce some important parameters and configurations for the proposed model or subsequent experiments. These settings are fixed in the whole process.



Class ID	Color	Class Name	Training	Testing	Total
1		Alfalfa	33	13	46
2		Corn-notill	100	1328	1428
3		Corn-mintill	100	730	830
4		Corn	100	137	237
5		Grass-pasture	100	383	483
6		Grass-trees	100	630	730
7		Grass-pasture-mowed	20	8	28
8		Hay-windrowed	100	378	478
9		Oats	14	6	20
10		Soybean-notill	100	872	972
11		Soybean-mintill	100	2355	2455
12		Soybean-clean	100	493	593
13		Wheat	100	105	205
14		Woods	100	1165	1265
15		Buildings-Grass-Trees-Drives	100	286	386
16		Stone-Steel-Towers	75	18	93
Total			1342	8907	10249
1		Asphalt	100	6531	6631
2		Meadows	100	18549	18649
3		Gravel	100	1999	2099
4		Trees	100	2964	3064
5		Painted metal sheets	100	1245	1345
6		Bare Soil	100	4929	5029
7		Bitumen	100	1230	1330
8		Self-Blocking Bricks	100	3582	3682
9		Shadows	100	847	947
Total			900	41876	42776
1		Scrub	33	728	761
2		Willow swamp	23	220	243
3		CP hammock	24	232	256
4		CP/Oak	24	228	252
5		Slash pine	15	146	161
6		Oak/Broadleaf	22	207	229
7		Hardwood swamp	9	96	105
8		Graminoid marsh	38	393	431
9		Spartina marsh	51	469	520
10		Cattail marsh	39	365	404
11		Salt marsh	41	378	419
12		Mud flats	49	454	503
13		Water	91	836	927
Total			459	4752	5211

Fig. 8. Detailed information of three data sets. (a) Indian Pines, (b) Pavia University, and (c) KSC, including false color map, corresponding ground truth, and setting of training and testing samples.

TABLE I
Setting of Scheme2 in SeMN

Layer	Kernel Number	Size	Stride	Padding	Activation
Conv	1	3	1	1	No
Max-Pooling	/	2	2	/	No
FC	128	128	/	/	ReLU

1) Setting of SeMN: Before fed into SeMN, original spectral vectors are all normalized to [−1, 1]. The number of neurons in hidden layers in all LSTMs is set to 128. In addition, 1-D CNN is used to obtain multiscale information in scheme2, whose implementation is listed in Table I. The stride is the step size of the sliding window of Conv and pooling layers in channel direction, while padding maintains the size of the 1-D feature map. Moreover, the kernel number and size of FC are all set as 128 because the FC layer can be regarded as an 1-D convolutional layer whose size equals to the number of neurons in input 1-D feature and the length of the output 1-D feature is the number of convolution kernels. The nonlinear mapping function ReLU is not placed on the convolutional layers since the convolution is only used to enlarge the receptive field.

2) Setting of SaMN: The basic setting of SaMN is almost the same as the MSCNN in [37]: the four components after PCA are also normalized to [−1, 1] and input into SaMN. In addition, the size of patches is set to 27 × 27 (28 × 28 in MSCNN) because we consider symmetry of the patch relative to the target center pixel, whereas a patch with a size of even number pixels cannot possess this characteristic.

Table II lists the configuration of different kinds of layers in SaMN. Dilation rate is the interval number between adjacent pixels when computing convolution (this value is 1 in standard convolutional layer). In addition, activation of FC is "ReLU/No" because the ReLU function is only applied on the FC layer after addition.

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TABLE II
SETTING OF SAMN

Layer	Kernel Number	Size	Stride	Padding	Dilation Rate	Activation
Conv	32	3×3	1	1	1	ReLU
Dilated Conv	32	3×3	1	2	2	ReLU
FC	128	/[1]	/	/	/	ReLU/No
ConvLSTM	32	3×3	1	1	1	No[2]

[1] Size of FC layer is mainly certained by the length of input feature. This parameter is uncertain because of the changed H/s, which will be discussed in next section.
[2] The ReLU function is embedded into ConvLSTM instead of activating its outputs.

3) Experimental Setting: Some important factors are fixed in all experiments. In SeMN, the learning rate is set to 1e-2, except at the situation of scheme2 when performing on Indian Pines data set, which is set to 1e-3. All learning rates in SaMN and ASSMN are set to 1e-2, and these learning rates are all decreased by half every 30 epochs. SeMN and SaMN are separately optimized through minibatch Adam and stochastic gradient descent strategy with momentum (SGDM) algorithms and parameters in LSTM parts are updated via backpropagation through time (BPTT) [61] algorithm. This arrangement takes advantage of the characteristics of fast convergence and higher precision in Adam and SGDM, respectively [62], because the former is more suitable to find a relatively better optimum in small amounts of data, whereas SGDM is used in SaMN for elaborate tuning so as to obtain a higher precision. In addition, the SGDM algorithm is also used in ASSMN because spatial part is of more importance in HSI spectral–spatial joint classification [37]. The training epoch is set to 200 with a batch size of 16. All experiments are repeatedly implemented ten times. Training samples are randomly selected in each trial and the remainder makes up testing set. The overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) are employed as the quantitative evaluation criteria, which are recorded with the mean value μ and standard deviation σ. All experiments are implemented on an Intel Xeon Sliver 4114 2.20-GHz processor and a single NVIDIA RTX 2080Ti GPU.

The experiments are divided into three parts. The first part is the analysis of SeMN, including scheme1 and scheme2. Then the evaluation of traditional and proposed strategy, plain and cascade style, influence of regularization in SaMN are discussed in the second part. In addition, parameter analyses of these two subnetworks are included in the first two parts. In the end, after a simple discussion of feature fusion, the performance comparison between the proposed networks and other state-of-the-art methods is presented.

C. Analysis of SeMN

1) Scheme1 Versus Scheme2: Considering band number in HSI, classification through scheme1 in SeMN is implemented by merging the LSTM features that source from the equally split eight parts, four parts, two parts, and one part (original) spectral vectors, is denoted as scheme1(8-4-2). In order to fairly compare these two schemes, three feature maps after pooling layer are also divided to eight, four and two parts

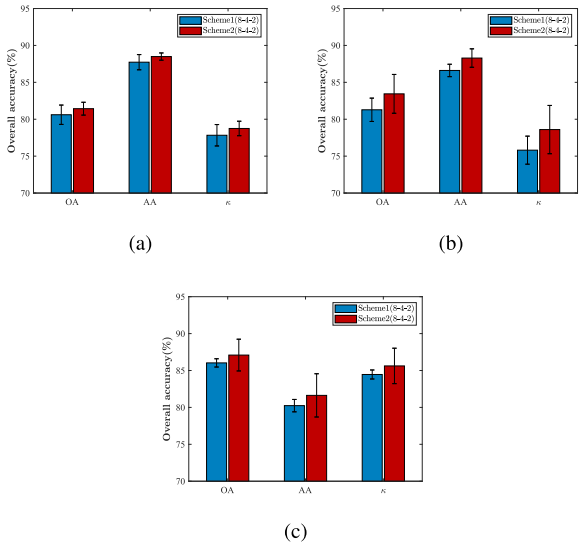

Fig. 9. Performance comparison between scheme1(8-4-2) and scheme2(8-4-2) on three data sets. (a) Indian Pines. (b) Pavia University. (c) KSC.

in turn and received by LSTMs. The information of original spectra identity is extracted by another LSTM and merged with the front three obtained features for classification. This solution is represented as scheme2(8-4-2).

Fig. 9 demonstrates the quantitative evaluation result of the above two schemes with fixed parameters. It is obvious that scheme2(8-4-2) is superior than scheme1(8-4-2) because the former network achieves better results in all metrics and all data sets since field vision of inputs at adjacent time steps exit intersection, meaning a closer distance in feature space, making it more convenient for LSTM to extract contextual information.

However, compared with scheme1(8-4-2), larger range of fluctuations are lying on scheme2(8-4-2), indicating the instability of the latter solution. These fluctuations are shown in Fig. 9 where black lines represent the upper and lower bounds of $\mu \pm \sigma$. This instability is particularly evident in Pavia University and KSC data sets.

Usually, it is cumbersome and unwise to configure the number of parts for each feature map in the grouping process. It is easier to set the same number for all features, meaning all LSTMs have same time steps. In order to investigate a powerful setting of scheme2, which is not only better than scheme2(8-4-2), but also more stable and simpler, we subsequently analyze the relationship between the number of time steps T_{se} and classification quality.

2) Parameter Analysis of Scheme2 in SeMN: Fig. 10 demonstrates the OA changes with different time steps T_{se} in scheme2 on three data sets. It can be seen that as the T_{se} increases, the change of OA's mean value μ is not obvious and has a small range of fluctuation on Indian Pines and KSC data sets, despite some improvement in early stage. While in Pavia University data set, OA gets decreased as the T_{se} increases. Therefore, the T_{se} in scheme2 is set to 2. In addition, the range of OA is relatively small when $T_{\text{se}} = 2$ compared with other time steps, meaning scheme2 is more stable under current setting, which is the most suitable

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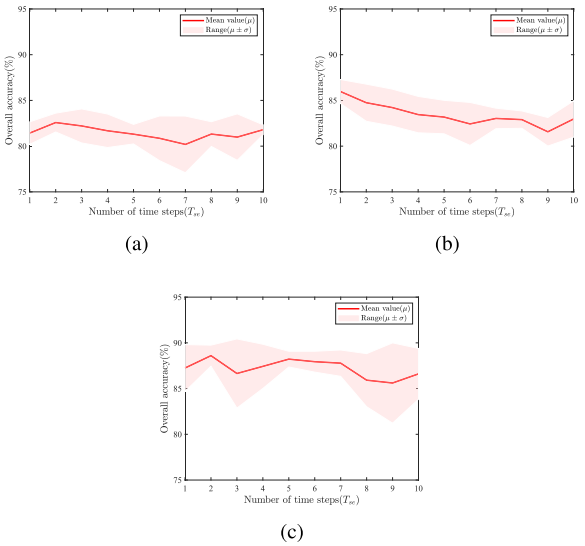

Fig. 10. Performance of scheme2 with different time steps on three data sets. (a) Indian Pines. (b) Pavia University. (c) KSC.

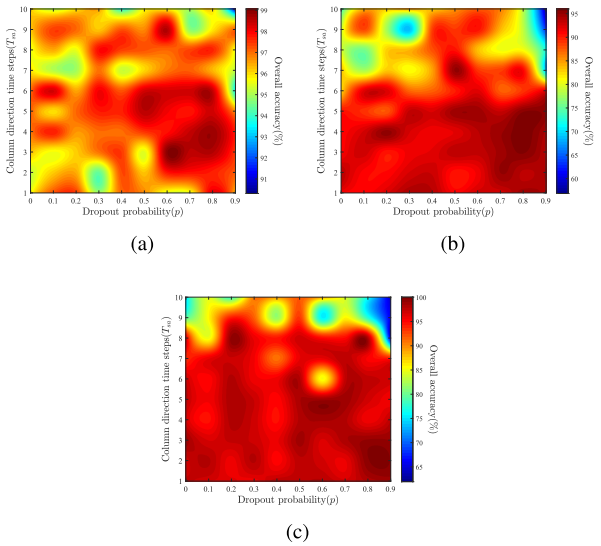

Fig. 11. Performance of SaMN with different column direction time steps and dropout probabilities on three data sets. (a) Indian Pines. (b) Pavia University. (c) KSC.

TABLE III
RESULT CHANGES OF SCHEME2(2-2-2) COMPARED WITH SCHEME2(8-4-2)

Improvement	(a)		(b)		(c)	
	μ	σ	μ	σ	μ	σ
OA(%)	+1.06	-0.01	+2.18	-0.61	+0.70	-0.32
AA(%)	+0.24	+0.81	+0.66	-0.31	+1.11	-0.31
$\kappa(\times 100)$	+1.19	-0.01	+2.69	-0.77	+0.80	-0.36

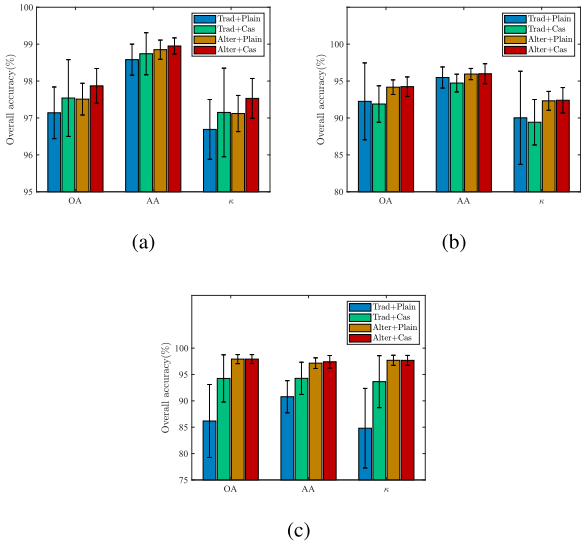

Fig. 12. Performance of SaMN with different spatial grouping and input sequence settings on three data sets. (a) Indian Pines. (b) Pavia University. (c) KSC.

setting for the aforementioned requirements. This model can be correspondingly represented as scheme2(2-2-2). Moreover, extra experiments are also manipulated to show the superiority of scheme2(2-2-2) compared with scheme2(8-4-2), and the results are shown in Table III.

Table III quantitatively describes the accuracy changes of scheme2(2-2-2) compared to scheme2(8-4-2) in OA, AA, and κ metrics on three data sets, where (a)–(c) represent Indian Pines, Pavia University, and KSC data sets respectively. It can be seen that almost all μs get increased and σs drop, indicating scheme2(2-2-2) can yield higher accuracies with stronger robustness. The scheme2(2-2-2) will be utilized as the spectral part of ASSMN in later experiments.

D. Analysis of SaMN

1) Parameter Estimation: There are two parameters H/s and p need to be adjusted in SaMN. H/s is the number of parts that obtained in row/column direction on each spatial patch, where H, s is the size of patch and split parts. H/s actually represents the number of time steps in ConvLSTM, and this parameter is denoted as T_{sa} for convenience. Another parameter p is the probability in dropout, which is the proportion of nonworking neurons in all neurons in previous FC layers before feature fusion, because too many or too few neurons can cause the phenomenon of overfitting or underfitting, and it is necessary to use the appropriate number of neurons for feature transformation.

Fig. 11 demonstrates the OA changes with different conditions of parameter T_{sa} and p, which are evaluated in the range of [1, 10] and [0, 0.9] with 1 and 0.1 intervals, respectively. Besides, the figures are smoothed with cubic interpolation in order to better show the impact of these parameters on model performance. It can be seen clearly that the high classification accuracy areas all locate in the bottom right parts in these figures. In top parts, the SaMN tends to be unstable and often has a poor performance, especially in the top right corner. In addition, the SaMN has a higher OA on Indian Pines data set compared with other two data sets. In this article, the T_{sa} and p are separately set to 3 and 0.8, and SaMN in this setting are utilized as the spatial part of ASSMN in later experiments.

2) Grouping Strategy and Input Sequence: Besides the estimation of T_{sa}, p, the proposed grouping strategy and

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TABLE IV
OA (%) OF DIFFERENT SETTINGS IN SAMN

Method	Indian Pines	Pavia University	KSC
backbone	95.72	84.42	89.23
backbone+ConvLSTM(no BN)	92.47	86.01	94.64
backbone+ConvLSTM+BN	96.53	90.82	96.25
backbone+ConvLSTM+dropout	95.17	85.61	96.04
backbone+ConvLSTM+BN+dropout	97.46	94.56	98.11

TABLE V
OA (%) OF DIFFERENT FEATURE COMBINING METHODS IN ASSMN

Method	Indian Pines	Pavia University	KSC
Feature concatenation	98.29	95.74	98.35
Feature weighted addition	98.17	95.80	97.88
Score weighted addition	98.30	96.26	98.44

input sequence setting are also tested and compared with the traditional grouping strategy and plain style. The results are shown in Fig. 12. The proposed alternate group strategy is depicted as "Alter" while traditional strategy is represented with "Trad". "Plain" and "Cas" separately represent the plain style and cascade style of input sequence setting.

Fig. 12 shows proposed strategy achieves better performance with higher mean values and smaller standard deviations compared with traditional strategy, especially in Pavia University and KSC data sets. These results verify that the model is more powerful and robust under the proposed alternate strategy, because current input of ConvLSTM in each time step can cover a larger range, making it more convenient for ConvLSTMs to extract contextual information from adjacent time steps. In addition, with the hierarchical contextual FE, the classification result is further improved when using cascade style of input sequence setting.

3) Ablation Study: To explore the contribution of proposed contextual information perception part in SaMN to the classification and the influence of regularization, a series of ablation experiments are implemented. The relative results are tabularized in Table IV. The "backbone" means the activated feature maps after dilated convolutional layers are flattened directly and fed into FC layers instead of the ConvLSTM layers.

The second row in Table IV indicates that simply employing ConvLSTM layers can certainly improve the classification result for Pavia University and KSC data sets. With the help of BN, the experiments results on all data sets get further improved, because BN well solves the overfitting problem which lies in ConvLSTM that contains a large amount of complex transformation. Dropout performs better when it is employed on the foundation of BN compared with using itself along.

E. Analysis of ASSMN and Performance Comparison

1) Feature Fusion Assessment: In this section, different feature combining methods are compared, and the corresponding results on these three data sets are shown in Table V. The experimental results prove our assumption that "feature concatenation" is superior to "feature weighted addition."

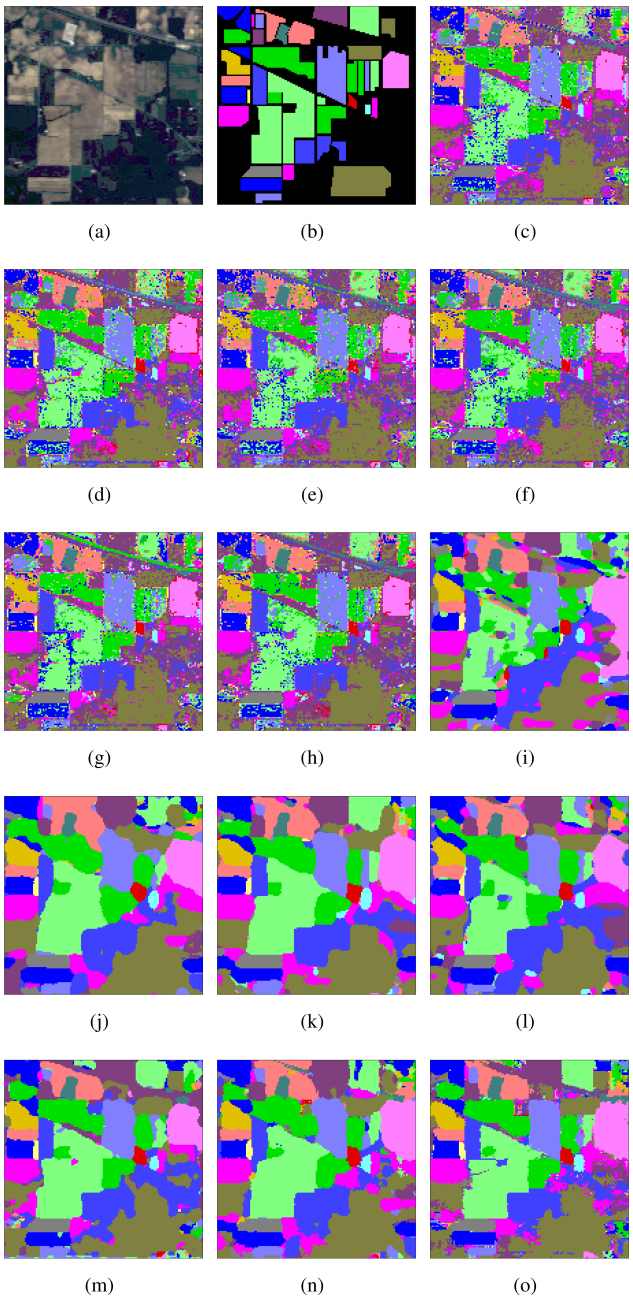

Fig. 13. Classification result maps on Indian Pines data set. (a) False color image. (b) Ground truth. (c) SVM. (d) 1-DCNN. (e) LSTM-strategy1. (f) LSTM-strategy2. (g) SeMN-scheme1. (h) SeMN-scheme2. (i) 2-DCNN. (j) MSCNN. (k) SaMN. (l) 3-DCNN. (m) SSRN. (n) SSUN. (o) ASSMN.

It can be observed that our method performs best on all three data sets with the strongest representation capability compared to other two methods, indicating the rationality and superiority of score fusion.

2) Classification Results: The proposed networks SeMN, SaMN, and ASSMN are simultaneously compared with several state-of-the-art deep learning methods, including 1-DCNN, 2-DCNN, 3-DCNN [39], LSTM-strategy1, LSTM-strategy2, MSCNN, SSUN [37], SSRN [38], and the two schemes in SeMN. Besides these deep learning methods, the SVM is also assessed as a representative in traditional methods.

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TABLE VI
EXPERIMENTAL DETAILS OF DIFFERENT ALGORITHMS ON INDIAN PINES DATA SET

Class	Spectral						Spatial			Joint			
Metric	SVM	1-DCNN	LSTM-strategy1	LSTM-strategy2	SeMN-scheme1	SeMN-scheme2	2-DCNN	MSCNN	SaMN	3-DCNN[1]	SSRN	SSUN	ASSMN
OA (%)	79.02	75.40	75.99	77.96	80.61	**82.49**	86.20	96.58	**97.46**	96.57	96.62	96.75	**98.30**
	±0.92	±4.18	±0.87	±2.04	±1.31	**±0.86**	±1.50	±0.79	**±0.52**	±1.12	±0.75	±0.65	**±0.51**
AA (%)	87.08	85.05	82.55	84.55	87.73	**88.73**	94.58	98.52	**98.85**	98.53	95.87	98.64	**99.09**
	±1.19	±1.77	±1.36	±1.74	±1.03	**±1.30**	±0.75	±0.30	**±0.16**	±0.41	±1.18	±0.23	**±0.36**
$\kappa \times 100$	76.04	72.06	72.63	74.81	77.83	**79.94**	84.20	96.05	**97.05**	96.04	96.09	96.24	**97.03**
	±1.04	±4.56	±0.99	±2.29	±1.45	**±0.97**	±1.69	±0.90	**±0.60**	±1.29	±0.86	±0.75	**±0.59**
T_{trn} (s)	**0.14**	255.91	119.45	118.95	177.19	218.75	**57.42**	100.70	1301.53	34686.45	869.76	**216.98**	1430.76
	±0.00	±3.16	±2.75	±3.16	±11.44	±2.64	**±2.80**	±5.57	±57.29	±1262.40	±31.28	**±3.28**	±43.07
T_{tes} (s)	**1.68**	0.46	2.05	1.90	0.53	0.52	**1.17**	1.83	5.14	144.42	4.02	**2.77**	4.75
	±0.02	±0.34	±0.83	±0.71	±0.07	±0.07	**±0.59**	±0.64	±1.34	±10.11	±0.33	**±0.69**	±1.07
1	94.62	89.23	81.54	87.69	92.31	90.00	100.00	100.00	100.00	100.00	100.00	100.00	99.23
2	75.02	72.97	69.98	72.15	74.74	77.68	82.27	95.86	94.33	94.65	95.26	95.74	96.48
3	76.51	74.01	73.51	75.66	76.90	76.03	92.92	97.16	98.55	97.30	96.96	97.55	98.68
4	90.00	84.31	84.31	85.62	88.25	89.05	99.64	99.85	100.00	100.00	95.20	100.00	99.71
5	92.98	90.99	89.90	88.83	92.09	93.52	93.58	98.62	99.37	98.98	98.88	97.96	98.88
6	95.11	94.14	91.16	92.57	94.54	95.62	96.52	99.02	99.37	99.57	98.51	99.67	99.97
7	90.00	95.00	88.75	90.00	91.25	90.00	100.00	100.00	100.00	100.00	95.71	100.00	98.75
8	97.86	97.94	95.34	95.48	97.96	98.28	99.63	99.95	100.00	100.00	100.00	100.00	100.00
9	90.00	85.00	73.33	81.67	95.00	93.33	100.00	100.00	100.00	100.00	100.00	100.00	100.00
10	79.64	78.97	79.79	81.69	85.38	87.02	89.39	95.67	96.59	96.66	87.79	95.92	98.20
11	64.64	59.71	63.49	68.06	69.54	72.41	69.85	93.46	96.41	93.88	99.01	93.71	97.29
12	84.89	82.76	75.88	83.57	84.91	87.81	96.31	98.50	97.81	98.34	96.59	99.23	99.23
13	98.95	99.14	98.48	96.29	98.19	99.24	99.90	100.00	100.00	100.00	99.89	100.00	99.52
14	88.87	77.48	85.05	82.11	87.12	88.44	93.93	98.55	99.23	97.76	99.54	98.52	99.45
15	76.40	79.65	73.04	71.92	78.85	81.33	99.37	99.72	100.00	99.97	94.10	100.00	100.00
16	97.78	99.44	97.22	99.44	96.67	100.00	100.00	100.00	100.00	99.44	76.45	100.00	100.00

[1] Running on three GPUs because of the high memory cost according to the setting in [39] (only for Indian Pines dataset).

TABLE VII
EXPERIMENTAL DETAILS OF DIFFERENT ALGORITHMS ON PAVIA UNIVERSITY DATA SET

Class	Spectral						Spatial			Joint			
Metric	SVM	1-DCNN	LSTM-strategy1	LSTM-strategy2	SeMN-scheme1	SeMN-scheme2	2-DCNN	MSCNN	SaMN	3-DCNN	SSRN	SSUN	ASSMN
OA (%)	**88.28**	75.06	71.46	67.49	81.28	85.62	71.67	94.26	**94.56**	92.98	96.17	95.36	**96.26**
	±0.61	±2.62	±6.71	±6.57	±1.58	±2.02	±12.12	±1.03	**±1.67**	±1.01	±3.27	±0.72	**±1.08**
AA (%)	**90.67**	83.65	80.96	72.52	86.61	88.95	80.76	96.01	**96.25**	95.13	96.30	97.09	**98.12**
	±0.56	±0.53	±1.86	±6.49	±0.84	±0.94	±4.01	±0.50	**±1.10**	±0.49	±2.91	±0.42	**±0.32**
$\kappa \times 100$	**84.65**	68.44	64.35	58.97	75.81	81.29	64.72	92.45	**92.82**	90.76	95.00	93.89	**95.06**
	±0.79	±2.74	±6.71	±7.54	±1.90	±2.50	±12.43	±1.32	**±2.15**	±1.28	±4.21	±0.93	**±1.40**
T_{trn} (s)	**0.03**	58.64	87.60	86.27	126.42	167.19	**46.92**	55.93	892.33	2602.84	579.45	**157.72**	976.86
	±0.00	±3.38	±2.92	±3.37	±5.57	±6.60	**±1.54**	±6.29	±20.92	±164.34	±13.06	**±4.41**	±22.31
T_{tes} (s)	**1.66**	1.68	4.98	4.34	2.61	2.92	**1.61**	3.26	21.43	161.40	21.02	**8.37**	20.21
	±0.06	±0.76	±0.96	±0.28	±0.20	±0.26	**±0.82**	±0.67	±2.19	±8.13	±0.78	**±0.86**	±1.99
1	81.11	72.81	71.33	76.99	81.32	84.53	70.94	91.52	94.22	90.16	99.21	93.73	96.80
2	88.83	68.73	63.40	63.97	78.42	84.14	66.66	92.85	93.45	92.23	99.61	93.74	94.06
3	82.85	78.76	69.16	55.98	80.33	81.40	75.10	93.23	91.51	92.62	95.11	94.33	97.95
4	95.58	89.89	86.45	91.82	94.13	94.07	89.02	97.49	98.49	96.39	89.56	98.49	99.21
5	99.37	99.65	99.33	99.51	98.75	99.04	99.97	99.54	99.97	99.94	99.99	100.00	100.00
6	89.57	75.44	72.70	63.29	75.96	84.96	51.23	98.05	93.03	91.63	92.97	98.53	97.92
7	93.49	90.39	87.89	55.67	91.42	90.89	84.74	97.77	99.33	97.88	97.77	98.96	99.54
8	85.31	77.46	78.64	45.85	79.24	81.68	90.07	95.25	96.81	95.57	92.57	96.56	97.68
9	99.95	99.74	99.85	99.63	99.93	99.81	99.08	98.41	99.47	99.75	99.95	99.47	99.94

We implement the SVM classification with the raw spectral vectors, which are normalized to $[-1, 1]$. In this article, the SVM is used with radial basis function kernel, and two important hyperparameters C and γ are certain through grid search with fivefold cross validation in the range of $\{2^{-5}, 2^{-4}, \ldots, 2^{19}\}$ and $\{2^{-15}, 2^{-14}, \ldots, 2^{5}\}$ [39]. This procedure is accomplished with LIBSVM [63]. The SSRN, SSUN, and its subnetworks (LSTM-strategy1, LSTM-strategy2, MSCNN) are operated with Tensorflow 1.14.0, while other deep learning methods are implemented on Pytorch 1.1.0.

From the category of input data and extracted features, these methods are divided into three groups. The pixelwise SVM, 1-DCNN, LSTM-strategy1, LSTM-strategy2, and our proposed SeMN-scheme1, SeMN-scheme2 are grouped in spectral part. 2-DCNN, MSCNN, and the proposed SaMN are divided into spatial group since they mainly perceive information in spatial domain around the target object. Other methods, which use 3-D convolution to extract spectral–spatial information simultaneously (3-DCNN, SSRN) or generate spectral–spatial features with the help of feature fusion (SSUN, ASSMN) consist the joint group.

Tables VI–VIII quantitatively demonstrate the OA, AA, κ, training time T_{trn}, testing time T_{tes} and accuracy of each category of these methods. The best results in each group are displayed in bold. It is obvious that spatial group performs better than spectral group with the information of surrounding objects, while joint group naturally achieves the best in each accuracy indicator on all data sets. In spectral group, our proposed methods perform better since the SeMN-scheme1 and

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TABLE VIII
EXPERIMENTAL DETAILS OF DIFFERENT ALGORITHMS ON KSC DATA SET

Class	Spectral						Spatial			Joint			
Metric	SVM	1-DCNN	LSTM-strategy1	LSTM-strategy2	SeMN-scheme1	SeMN-scheme2	2-DCNN	MSCNN	SaMN	3-DCNN	SSRN	SSUN	ASSMN
OA (%)	**90.84** ±**0.72**	78.44 ±2.54	78.88 ±6.18	77.86 ±4.45	86.03 ±0.56	87.79 ±1.83	93.05 ±0.77	97.75 ±0.68	**98.11** ±**0.62**	97.92 ±0.76	98.30 ±1.18	98.13 ±0.74	**98.44** ±**0.92**
AA (%)	**86.48** ±**0.68**	69.14 ±3.11	70.32 ±9.35	68.13 ±7.57	80.24 ±0.84	82.74 ±2.62	91.40 ±1.09	97.13 ±0.93	**97.49** ±**0.87**	97.31 ±1.09	97.51 ±1.47	97.80 ±0.67	**98.00** ±**1.03**
$\kappa \times 100$	**89.80** ±**0.79**	75.99 ±2.76	76.45 ±6.91	75.24 ±5.00	84.46 ±0.61	86.42 ±2.04	92.26 ±0.86	97.49 ±0.76	**97.90** ±**0.69**	97.68 ±0.84	98.11 ±1.31	97.91 ±0.82	**98.27** ±**1.02**
T_{trn} (s)	**0.02** ±**0.00**	68.18 ±2.27	49.27 ±3.00	49.28 ±4.45	64.14 ±3.79	81.69 ±4.34	**24.25** ±**1.17**	40.30 ±5.15	391.65 ±3.38	4616.92 ±204.81	332.95 ±11.31	**85.85** ±**4.74**	472.07 ±7.84
T_{tes} (s)	**0.17** ±**0.01**	1.03 ±0.51	1.42 ±0.70	1.32 ±0.55	0.30 ±0.05	0.33 ±0.10	**0.98** ±**0.46**	1.15 ±0.45	3.05 ±1.48	90.30 ±8.19	2.48 ±0.18	**1.96** ±**0.35**	2.27 ±0.30
1	89.73	79.59	83.54	88.01	83.05	84.23	86.15	95.67	96.24	95.71	99.67	95.71	97.09
2	89.14	87.73	72.50	83.59	84.14	88.00	92.68	93.73	95.95	95.91	98.91	96.14	96.91
3	91.68	37.89	56.77	35.39	83.49	86.85	92.33	96.21	95.60	93.58	98.02	95.95	93.88
4	78.07	29.69	46.89	29.04	56.01	63.11	78.60	90.44	93.55	93.42	86.92	93.11	93.25
5	56.85	41.37	40.55	44.93	51.03	58.97	96.44	97.53	93.70	98.08	96.66	98.15	98.15
6	60.82	40.00	37.44	40.19	51.55	53.04	73.33	97.58	97.68	95.27	94.97	98.60	97.73
7	83.85	51.67	59.38	53.44	74.79	76.25	86.04	95.94	97.60	95.83	94.76	97.81	98.85
8	90.59	70.66	59.11	53.79	84.86	85.06	94.78	97.53	98.07	98.93	98.81	97.99	98.98
9	95.95	87.33	79.53	81.60	90.55	93.35	97.10	99.64	99.70	98.83	99.86	99.06	99.72
10	97.10	92.49	93.62	91.34	94.77	96.11	98.58	99.81	99.92	99.92	99.73	99.21	99.97
11	96.64	95.56	97.04	96.85	96.56	96.88	97.49	100.00	100.00	100.00	99.62	100.00	100.00
12	93.79	84.87	87.80	87.51	92.58	93.79	94.69	98.55	99.38	99.49	99.78	99.71	99.47
13	100.00	99.92	99.99	100.00	99.93	99.95	100.00	100.00	100.00	100.00	99.90	100.00	100.00

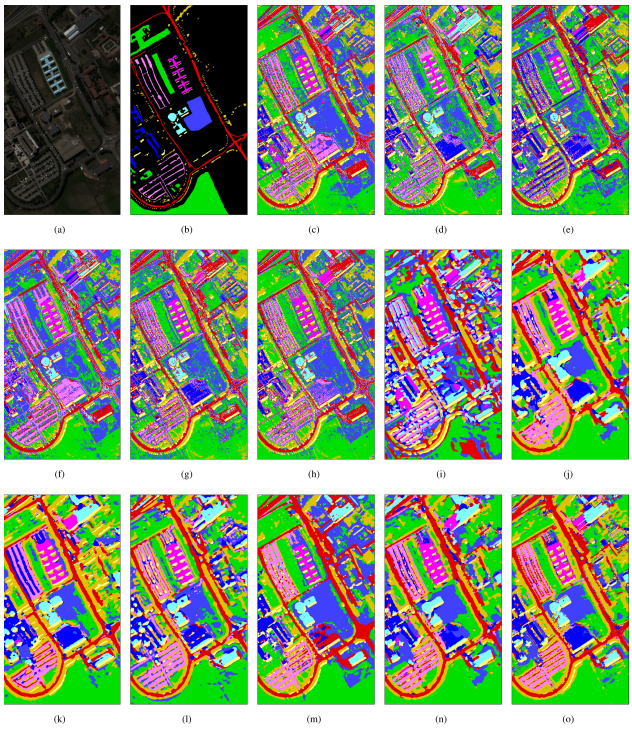

Fig. 14. Classification result maps on Pavia University data set. (a) False color image. (b) Ground truth. (c) SVM. (d) 1-DCNN. (e) LSTM-strategy1. (f) LSTM-strategy2. (g) SeMN-scheme1. (h) SeMN-scheme2. (i) 2-DCNN. (j) MSCNN. (k) SaMN. (l) 3-DCNN. (m) SSRN. (n) SSUN. (o) ASSMN.

SeMN-scheme2 fully use the multiscale information compared to other single-scale FE networks such as 1-DCNN and two grouping strategy LSTMs. In addition, with a closer distance in feature space and more natural transition of adjacent time step inputs, it is more convenient for LSTM to extract contextual information. Thus, the SeMN-secheme2 further outperforms SeMN-scheme1. It is worth noting that the SVM achieves the best at all metrics on Pavia University and KSC data set in spectral group, meaning the SVM can achieve relatively high accuracy in a shorter period of time, indicating the merits of traditional method. But the SVM is difficult to be as a feature extractor to obtain spectral feature which can be used to fuse with the extracted spatial feature. In the spatial group, under the aid of designed alternate grouping strategy, cascade input sequence setting and regularization, the ConvLSTM successfully extract spatial contextual information and the feature is obtained from multiple scales. Therefore, our SaMN performs better compared with other methods. The shortest running time is achieve by 2-DCNN, however, since only one principal component is used as input [39], its performance is not satisfactory, especially on Pavia University data set. In joint group, all methods perform well and the proposed ASSMN achieves the best on all data sets. From the aspects of operating time, the 3-D convolution-based methods tend to need more computation and more time. Note that there are many nonlinear transformations or activation functions in ConvLSTM, which limits the efficiency of ASSMN, and the shortest running time is achieved by SSUN.

Figs. 13–15 show the classification maps of different methods along with the corresponding ground truth maps and original images. It is obvious that the spectral classification methods tend to generate strong salt-and-pepper noise, causing the rough surfaces inside objects since these methods only consider the spectral characteristic of the target pixel and ignore the surrounding environment. On the other hand, these methods usually possess a clear outline and retain the shape of objects since the classification is not interfered by other pixels. This phenomenon is especially obvious in Indian Pines and Pavia University data sets. With the spatial information contained in surrounding pixels, the pepper noise is suppressed effectively and more smooth and clean maps are obtained by spatial classification methods. Compared with the best results in spectral group, the accuracy of the proposed SaMN improves by 14.97%, 6.28%, and 7.27% in three different

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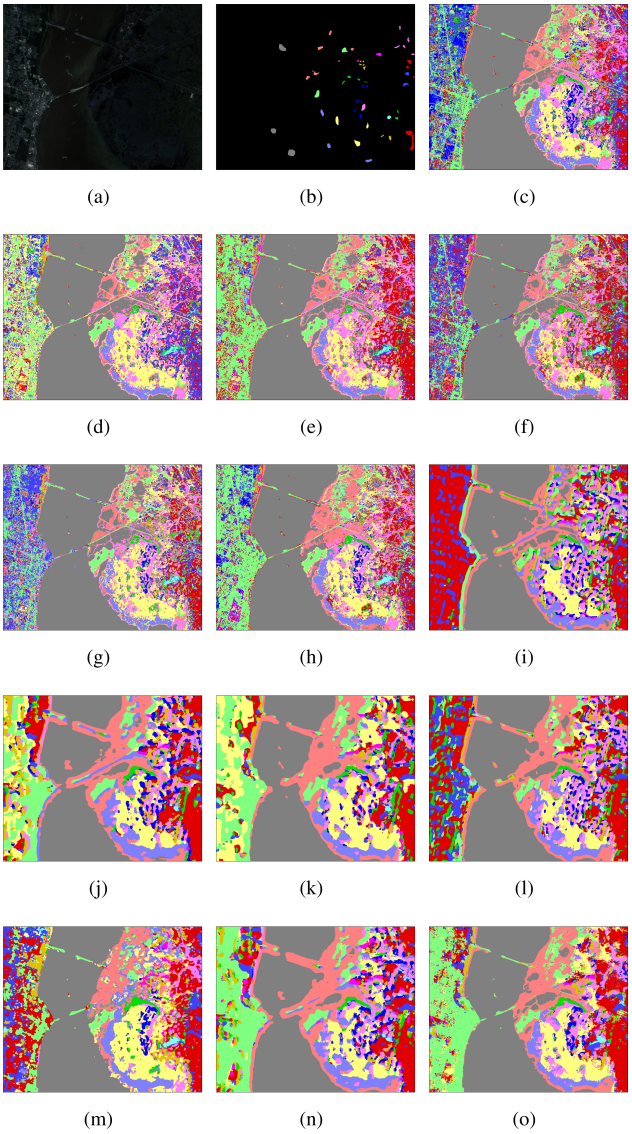

Fig. 15. Classification result maps on KSC data set. (a) False color image. (b) Ground truth. (c) SVM. (d) 1-DCNN. (e) LSTM-strategy1. (f) LSTM-strategy2. (g) SeMN-scheme1. (h) SeMN-scheme2. (i) 2-DCNN. (j) MSCNN. (k) SaMN. (l) 3-DCNN. (m) SSRN. (n) SSUN. (o) ASSMN.

data sets respectively. However, these 2-D convolution-based spatial FE methods often suffer from the challenge of over-smoothing that the small objects are seriously interfered or even misclassified because of the large receptive field in the extracted spatial features. For example, the narrow road in the upper part of Indian Pines data and many tiny and isolate "CP hammocks" in KSC data vanish in MSCNN and the proposed SaMN. According to the above discussion, in order to obtain a classification map where adjacent objects and pure targets can be clearly distinguished, it is necessary to simultaneously use spectral and spatial features. Benefitting from both characteristics, the methods in joint group obtain a higher accuracy and produce the best classification maps. The proposed ASSMN better maintains the edge of the targets which can be seen from the discriminative edges and corners, especially the purple and green objects in the center part of Indian Pines image compared with 3-DCNN, SSRN, and SSUN. In Pavia University data set, the ASSMN overcomes

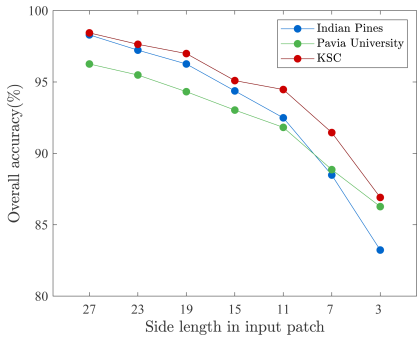

Fig. 16. Performance of ASSMN with different sizes input patches.

TABLE IX
MODEL COMPLEXITY OF THE PROPOSED NETWORKS

Model complexity	SeMN-scheme1	SeMN-scheme2	SaMN	ASSMN
#Params(k)	476.82	431.77	1951.06	2382.83
FLOPs(k)	945.15	857.14	1065245.18	1066102.32

the oversmoothing problem which happens in 3-DCNN and SSUN. It retains many small objects and generates a satisfactory classification map.

3) Influence of Input Size: Besides 27×27, the performance of ASSMN with smaller size inputs (23×23, 19×19, 15×15, 11×11, 7×7 and 3×3) are also evaluated, and the results are shown in Fig. 16. It can be seen that the classification accuracies of all data sets decrease when reducing the input patch sizes. Nevertheless, the OAs also exceed 90 even if the side length is reduced by half. If the size of patches is set to 3×3, the results are close to the accuracy of SeMN shown in Tables VI–VIII. These two observations indicate that the ConvLSTM can play a better role on larger patch because large-scale patches can help ConvLSTM fully capture the spatial contextual information, so as to reach or exceed the level of existing advanced methods.

4) Model Complexity: Table IX depicts the complexity of the proposed networks. Specifically, the number of parameters (#Params) and floating-point operations per second (FLOPs) are evaluated. Compared with SeMN, with the larger input patch and ConvLSTM of multi time step calculation, the complexity in SaMN and ASSMN is much higher. In particular, due to the parameters in ConvLSTM are continuously involved into repeated calculations, compared with SeMN, the #Params in SaMN and ASSMN increases less than ten times while the FLOPs have grown more than a thousand times. For example, a convolution operation in ConvLSTM requires about 1e4 kFLOPs.

VII. CONCLUSION

In this article, we propose an end-to-end adaptive classification network ASSMN to extract multiscale contextual information for HSI classification, which contains spectral FE and spatial FE subnetworks (SeMN and SaMN). Compared with previous works that spectral information is obtained in a single scale, two multiscale schemes are developed based on

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wavelength or vision field in SeMN with the help of band grouping strategy. In SaMN, the dilated convolution layer and activation layer are used to substitute max-pooling layer. This arrangement simultaneously increases the receptive field and preserves the resolution of feature map. On the foundation of existing multiscale architecture, the spatial contextual information is obtained with the help of ConvLSTM, and the multidimensional alternate grouping strategy makes the inputs of different time steps have better communication. Another design, the cascade style input sequence setting balance the computational efficiency and contextual information perceiving in row and column directions simultaneously. After obtaining the features from SeMN and SaMN, an adaptive feature combining module called "score weighted" is proposed, when considering the different importance of spectral and spatial parts.

Experiments on Indian Pines, Pavia University, and KSC data sets show that our methods benefit from both characteristics of spectral and spatial features, obtaining a higher accuracy and producing classification maps with discriminative edges and continuous surfaces between or inside objects compared with other state-of-the-art methods. In addition, this article also proves the effectiveness of regularization when processing the overfitting problem in ConvLSTM.

In Tables VI–VIII, it can be seen that our proposed methods are still time-consuming. In fact, it is a common shortage in pixel-level deep learning-based classification methods because of the large overlap in adjacent input patches. Therefore, our future work may explore the HSI classification at image or object level using fully convolution networks to further improve the classification efficiency.

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